Homework 3.

Problem 1

Proof of $E[x - \hat{x}]$

$$E[x - \hat{x}] = \sum_{i=0}^{M-1} \int_{t_i}^{t_{i+1}} (x - \hat{x}_i) p(x)\, dx \quad x \in X \quad \hat{x}_i \in \hat{X}_i$$

$$= \sum \left(\int_{t_i}^{t_{i+1}} x\, p(x)\, dx - \int_{t_i}^{t_{i+1}} \hat{x}_i p(x)\, dx \right)$$

由积分定理.

$$\hat{x}_i \int_{t_i}^{t_{i+1}} p(x)\, dx = \int_{t_i}^{t_{i+1}} x\, p(x)\, dx \quad \Rightarrow \quad E[(x - \hat{x}_i)] = 0 .$$

Proof $E[(x - \hat{x})\hat{x}] = 0$.

$$E[(x - \hat{x})\hat{x}] = \sum_{i=0}^{M-1} \int_{t_i}^{t_{i-1}} (x_i - \hat{x}_i)\hat{x}_i\, p(x)\, dx$$

$$= \sum \hat{x}_i \underbrace{\int_{t_i}^{t_{i-1}} (x_i - \hat{x}_i) p(x)\, dx}_{0 .}$$

$$= 0 .$$

$$\sigma_{\hat{x}}^2 = \sigma_x^2 + E[(x - \hat{x})]^2 = 0$$

$$\because E[(x - \hat{x})] = 0 \quad \Rightarrow \quad E[x] = E[\hat{x}]$$

$$\sigma_{\hat{x}}^2 = \sigma_x^2 + E[(x - \hat{x})^2]$$

$$\Rightarrow E[(\hat{x} - E[\hat{x}])^2] = E[(x - E[x])^2] + E[(x - \hat{x})^2] \quad (1)$$

$$X = \hat{x} + (x - \hat{x})$$

$$x - E(x) = (\hat{x} - E(\hat{x})) + (x - \hat{x})$$

$$(x - E(x))^2 = [\hat{x} - E(\hat{x})]^2 + (x - \hat{x})^2 + 2(\hat{x} - E(\hat{x}))(x - \hat{x})$$

$$E[(x - E(x))^2] = E[(\hat{x} - E(\hat{x}))^2] + E[(x - \hat{x})^2] + \underbrace{2E[(\hat{x} - E(\hat{x}))(x - \hat{x})]}_{②}$$

若想证明式①，证明②=0即可

对于② $E[(\hat{x} - E(\hat{x}))(x - \hat{x})]$

$$= E[\hat{x}\underbrace{(x - \hat{x})}_{0}] - E[E(\hat{x})\underbrace{(x - \hat{x})}_{0}] = 0.$$

综上 $E[(x - E(x))^2] = E[(\hat{x} - E(\hat{x}))^2] + E[(x - \hat{x})^2]$.

$$\Rightarrow \quad \sigma_x^2 = \sigma_{\hat{x}}^2 + E[(x - \hat{x})^2].$$

Problem 2.

distortion $d = \|x - \hat{x}\|_2^2$.

要证六边形能有更好的 distortion 即证六边形有最小的平均失真，

由于 X_1, X_2 均匀分布，又因为量化误差实际是衡量量化区域内一个点到中心点（重心点）的距离.

原问题可以转化为 相同面积下，六边形比正方形的平均失真小

假设 M 个量化区间 在 $[a,b] \times [a,b]$ 内

则小区域面积 $S = \frac{(b-a)^2}{M}$

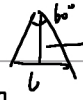
对于边长 l 的正方形，$S = l^2$.

$$D_{square} = \frac{1}{S}\iint (x^2+y^2)\,dx\,dy \quad -\frac{l}{2} \le x \le \frac{l}{2},\ -\frac{l}{2} \le y \le \frac{l}{2}.$$

$$\Rightarrow \frac{1}{l^2}\int_{-\frac{l}{2}}^{\frac{l}{2}}\int_{-\frac{l}{2}}^{\frac{l}{2}} (x^2+y^2)\,dx\,dy.$$

$$= \frac{l^2}{6} = \frac{S}{6}.$$

对于边长 l 的正六边形 $S = \frac{3\sqrt{3}}{2}l^2$



$\Rightarrow a = \frac{\sqrt{3}}{2}l.$

$r\cos\theta = a.$

$r(\theta) = \frac{\sqrt{3}\,l}{2\cos\theta}.$

$\theta \in [-\frac{\pi}{6}, \frac{\pi}{6}]$

$$6 \times \int_{-\frac{\pi}{6}}^{\frac{\pi}{6}}\int_0^{\frac{\sqrt{3}\,l}{2\cos\theta}} r^2 \cdot r\,dr\,d\theta.$$

$$D_{六} = \frac{1}{S} \times 6 \times \frac{9l^2}{64} \cdot \frac{2}{9\sqrt{3}} = \frac{\frac{5\sqrt{3}}{8}l^4}{\frac{3\sqrt{3}\,l^2}{2}} = \frac{5}{12}l^2$$

又 $l^2 = \frac{2S}{3\sqrt{3}}$ $\quad D_{六} = \frac{5S}{18\sqrt{3}}$ $\quad \frac{5}{18\sqrt{3}} < \frac{1}{6}$

因此 $D_{六} < D_{square}$. 证毕.